

SRM University – AP, Andhra Pradesh
Neeru Konda, Mangalagiri Mandal
Guntur District, Mangalagiri, Andhra Pradesh 522240

Name of the Course: Blockchain Technologies

Course Code	AML 571	Course Category	Core Elective (CW)	L-T-P-C	3	0	0	3
Pre-Requisite Course(s)		Co-Requisite Course(s)		Progressive Course(s)				
Course Offering Department	CSE	Professional / Licensing Standards						

Course Objectives / Course Learning Rationales (CLRs)

Objective 1. Learn the cryptographic and distributed-systems foundations of blockchain technology and understand the architecture of public ledgers.

Objective 2. To introduce the fundamentals of consensus algorithms, smart contracts, and the Ethereum Virtual Machine execution model.

Objective 3. To discuss various issues including scalability, privacy, security, and governance that make blockchain a hard engineering problem.

Objective 4. To discuss well-known applications of blockchain such as DeFi, Zero-Knowledge Proofs, Rollups, and On-Chain Machine Learning.

Course Outcomes / Course Learning Outcomes (CLOs)

	At the end of the course the learner will be able to	Bloom's Level	Expected Proficiency Percentage	Expected Attainment Percentage
Outcome 1	Recall the fundamental concepts of cryptographic primitives and distributed consensus.	1	70%	68%
Outcome 2	Demonstrate algorithms for blockchain consensus, digital signatures, and Merkle structures.	2	70%	65%
Outcome 3	Develop smart contracts and Layer-2 systems for decentralized financial applications.	3	70%	60%
Outcome 4	Implement systems using Zero-Knowledge Proofs and analyze DeFi/security incident postmortems.	4	70%	65%

Course Articulation Matrix (CLO) to Program Learning Outcomes (PLO)

CLOs	Eng ineer ing Kno wle dge	Desi gn / Dev elop men t of Solu tion s	Con duct Inve stig ations of C omp lex Pro ble ms	Con duct Inve stig ations of C omp lex Pro ble ms	Mo der n T ool Usa ge	The Eng ineer and Soci ety	Env iron men t and Sust aina bilit y	Ethi cs	Indi vidu al and Tea m Wo rk	Co mmu nic ation	Life -lon g Le arning	PS O 1	PS O 2	PS O 3
Outco me 1	2	3	3		2							3	2	
Outco me 2	2	3	3		2							2	2	
Outco me 3	2	3	2		2							2	2	
Outco me 4	3	3	3		2							2	3	
Course Avera ge	2	3	3		2							2	2	

Course Utilization Plan - Theory

Unit No.	Unit Name	Required Contact Hours	CLOs Addressed	References Used
Unit 1		9		
	Introduction and Overview: What is Blockchain; Why decentralization is hard: Byzantine faults, double-spend; Applications of Blockchain.	2	1	1,2
	Cryptographic Foundations: Hash functions (SHA-256, Keccak), pre-image / collision resistance, puzzle-friendliness.	2	1	1,3
	Merkle Trees and Patricia Tries; Inclusion proofs; Verkle trees and KZG polynomial commitments.	2	1	1,3
	Digital Signatures: ECDSA, EdDSA, Schnorr (BIP-340), BLS aggregate signatures.	2	1	1,3
	Distributed systems: CAP / FLP impossibility, Byzantine Generals Problem, the 3f+1 bound.	1	1,2	1,2
Unit 2		8		
	Bitcoin Deep Dive: UTXO model, Bitcoin Script, Proof-of-Work, difficulty adjustment, longest-chain rule.	2	1,2	1,4
	Bitcoin economics: block reward halving, fees, selfish mining (Eyal & Sirer), 51% attack.	1	1,2	1,4
	Ethereum & Smart Contracts: Account model, EVM, gas mechanics, Solidity language fundamentals.	2	1,2,3	1,5
	ERC standards (ERC-20, ERC-721, ERC-4337); Account abstraction; client diversity.	2	2,3	5,6
	Lightning Network and payment channels; Off-chain state.	1	2,3	1,6
Unit 3		7		
	Consensus Algorithms — Proof of Work variants; Energy and centralization tradeoffs.	1	1,2	1,7
	Proof of Stake families: Casper FFG + LMD-GHOST (Ethereum), Tendermint / CometBFT, Ouroboros (Cardano).	2	1,2	7,8
	PBFT and HotStuff: 3-phase voting, linear view change, pipelined finality.	2	1,2	7

Unit No.	Unit Name	Required Contact Hours	CLOs Addressed	References Used
	Algorand BA*, Avalanche Snowball, Solana PoH; DAG protocols (Narwhal, Bullshark).	1	1,2	7,8
	Slashing conditions, nothing-at-stake problem, long-range attacks, weak subjectivity.	1	2,3	7,8
Unit 4		9		
	Scalability and the Trilemma; Sharding; Vertical vs Horizontal scaling.	2	2,3	5,6
	Layer 2: Optimistic Rollups (Arbitrum, Optimism) and fraud proof bisection.	2	3	5,6,9
	ZK Rollups (zkSync, StarkNet, Polygon zkEVM); zkEVM equivalence levels (Type 1–4).	2	3,4	5,6,9
	Data Availability: EIP-4844 proto-danksharding, blob transactions, Celestia, EigenDA.	1	3,4	5,6
	Zero-Knowledge Proofs: SNARK vs STARK, Groth16, PLONK, Halo2; Circom toolchain.	2	3,4	9,10
Unit 5		12		
	Decentralized Finance: Stablecoins (USDC, DAI, FRAX), CDPs, algorithmic failure (Terra/UST).	2	3,4	5,6
	Automated Market Makers: Uniswap v2 (xy=k), Curve StableSwap, Uniswap v3 concentrated liquidity.	2	3,4	5,6
	Lending markets (Aave, Compound, Morpho); Flash loans; Oracle manipulation.	2	3,4	5,6
	MEV: arbitrage, sandwich attacks, MEV-Boost, Proposer-Builder Separation (PBS).	1	3,4	5,6
	Security: Reentrancy, integer overflow, access control; DAO hack, Wormhole, Ronin, Curve postmortems.	2	4	5,6,11
	Governance and Tokenomics: DAOs, on-chain voting, veToken model, restaking (EigenLayer).	2	4	5,6
	Research Frontiers: zkML, ZKVMs (RISC Zero, SP1), folding schemes (Nova, HyperNova), modular blockchains.	1	4	9,10,12
	Total Contact Hours	45		

Course Utilization Plan – Suggested In-Class Projects

Exp No.	Experiment Name	CLOs Addressed	References Used
1	Build a toy blockchain in Python with hash chain and tamper detection.	1,2	1,3
2	Implement ECDSA wallet, transaction signing and signature verification.	1,2	1,3
3	Construct Merkle tree and verify inclusion proofs for a block of transactions.	1,2	1,3
4	Send a Bitcoin testnet transaction and decode raw transaction bytes.	1,2	1,4
5	Develop and deploy an ERC-20 token contract using Foundry / Solidity.	2,3	5,6
6	Reproduce a reentrancy attack on a vulnerable contract; apply checks-effects-interactions fix.	3,4	5,6,11
7	Write invariant tests in Foundry; integrate Slither static analysis.	3,4	5,6,11
8	Build a Proof-of-Work vs Proof-of-Stake simulator and benchmark finality latency.	2,3	7,8
9	Inspect an Optimistic Rollup batch on Etherscan; compute calldata gas savings.	3,4	5,6
10	Write a ZK circuit in Circom proving knowledge of x with $x^2=25$; deploy verifier.	3,4	9,10
11	Analyze EIP-4844 blob transactions; compute cost-per-byte savings over calldata.	3,4	5,6
12	Implement a Uniswap-v2-style AMM and compute impermanent loss for price moves.	3,4	5,6
13	Simulate a flash loan arbitrage across two DEX pools using a mainnet fork.	4	5,6
14	Implement a DAO voting contract with token-weighted and conviction voting.	3,4	5,6
15	Build a zkML inference proof for a small MLP using EZKL; verify on Sepolia.	4	9,10,12

Recommended Resources

1. Narayanan, A., Bonneau, J., Felten, E., Miller, A., Goldfeder, S. (2016). Bitcoin and Cryptocurrency Technologies. Princeton University Press.
2. Antonopoulos, A. M. (2017). Mastering Bitcoin: Programming the Open Blockchain (2nd ed). O'Reilly.
3. Boneh, D., Shoup, V. (2023). A Graduate Course in Applied Cryptography (online).
4. Antonopoulos, A. M., Wood, G. (2018). Mastering Ethereum: Building Smart Contracts and DApps. O'Reilly.
5. Werner, S. et al. (2022). SoK: Decentralized Finance (DeFi). ACM AFT.
6. <https://ethereum.org/en/developers/docs/> (Ethereum developer documentation)
7. Yin, M. et al. (2019). HotStuff: BFT Consensus with Linearity and Responsiveness. ACM PODC.
8. Kiayias, A. et al. (2017). Ouroboros: A Provably Secure Proof-of-Stake Protocol. CRYPTO.
9. Ben-Sasson, E. et al. (2018). Scalable, transparent, and post-quantum secure computational integrity (STARK).
10. Gabizon, A., Williamson, Z., Ciobotaru, O. (2019). PLONK: Permutations over Lagrange-bases.
11. Atzei, N., Bartoletti, M., Cimoli, T. (2017). A Survey of Attacks on Ethereum Smart Contracts. POST.
12. <https://web.stanford.edu/class/cs251/> (Stanford CS251 — Cryptocurrencies and Blockchain Technologies)

Other Resources

1. <https://bitcoin.org/bitcoin.pdf> (Satoshi Nakamoto, 2008)
2. <https://vitalik.eth.limo/> (Vitalik Buterin's research notes)
3. <https://ethresear.ch/> (Ethereum Research Forum)
4. <https://github.com/foundry-rs/foundry> (Foundry development toolkit)
5. <https://zk-learning.org/> (ZK MOOC — Stanford / Berkeley)
6. <https://learn.cyfrin.io/> (Cyfrin Updraft — Smart Contract Developer track)

Learning Assessment

Bloom's Level of Cognitive Task	Continuous Learning Assessments (50%)								End Semester Exam (50%)	
	CLA-1 (10%)		Mid-1 (15%)		CLA-2 (10%)		Mid-2 (15%)			
	Th	Prac	Th	Prac	Th	Prac	Th	Prac	Th	Prac
Level 1 Remember / Understand	70%		65%		40%		50%		40%	
Level 2 Apply / Analyse	30%		35%		40%		50%		60%	

Level 3 Evaluate / Create					20%					
Total	100%		100%		100%		100%		100%	